Student guide | Lesson 3 of 5

Food

Random food appears and the snake eats it.

Name: _____ Date: _____

Today you will make

A red dot appears at a random place. When the head touches it, the dot jumps somewhere else. That is your snake eating.

Words you will meet

Word	What it means
random	A module that picks numbers by chance.
randrange	Picks a random number in a range, counting in steps.
distance()	How far one turtle is from another.
constant	A variable we never change while the game runs, like STEP. We write it in CAPITALS.
helper function	A small function that does one job for the rest of the game.

Before you start

- Open snake2.py and save it as **snake3.py**.
- If your Lesson 2 does not work, use the complete code at the end of the Lesson 2 guide.

Step 1: Import random

Add this on the line under import turtle:

```
import random
```

What it does

- This loads the toolbox that can pick random numbers.

Step 2: Make the food

Add this **after the head setup**:

```
food = turtle.Turtle("circle")
food.color("red")
food.penup()
food.goto(0, 100)
```

What it does

- It looks just like the head, but it is a red circle.
- **goto(0, 100)** puts it 100 pixels above the middle.

Try it: Run it. A red dot should appear above the green square.

Step 3: A function to move the food

Add this **above game_loop**:

```
def move_food():
    food.goto(random.randrange(-LIMIT, LIMIT, STEP),
              random.randrange(-LIMIT, LIMIT, STEP))
```

What it does

- **randrange(-280, 280, 20)** picks from -280, -260, -240, and so on up to 260.
- The third number (20) keeps the food on the same grid as the snake.
- We do this once for x and once for y, so the food can land anywhere.

> **Try it:** Test it in the Python shell: type import random, then random.randrange(-280, 280, 20) several times.

Step 4: Eat the food

Change game_loop so it looks like this. The new lines are marked.

```
def game_loop():
    if abs(head.xcor()) > LIMIT or abs(head.ycor()) > LIMIT:
        head.goto(0, 0)
        head.direction = "stop"

    if head.distance(food) < STEP:     # new
        move_food()                    # new

    move_head()
    screen.update()
    screen.ontimer(game_loop, 100)
```

What it does

- **head.distance(food)** is how far the head is from the food.
- If it is less than STEP, they are touching, so the food jumps to a new spot.

> **Try it:** Run it and steer onto the red dot. It should jump away every time.

Think about it: Why do we check distance < STEP instead of head and food being in the exact same place?

Your answer: _____

Challenges (pick one, or do them all)

[ ] Change the food's colour and shape.

[ ] Make the food only appear in the top half of the screen.

[ ] Change the eating distance from STEP to 40. How does it feel?

Stuck? Check here first

What you see	Likely cause and fix
NameError: name 'random' is not defined	import random is missing at the top.
The food never moves	The if block is not inside game_loop, or the indentation is wrong.
The food lands between grid squares	The third number (STEP) is missing from randrange.

The food is eaten from too far away	The distance number is too big. STEP (20) is about one square.

I can...

[ ] generate random numbers.

[ ] measure how close two turtles are.

[ ] explain why we use < instead of == for touching.

[ ] write a helper function and call it from the game loop.

Check your work: the complete code

Compare your file with this one, line by line. If yours does not run, the mistake is usually a typo, a missing colon, or indentation that is not lined up.

```
import turtle
import random

STEP = 20
LIMIT = 280

screen = turtle.Screen()
screen.title("Snake")
screen.bgcolor("black")
screen.setup(600, 600)
screen.tracer(0)

head = turtle.Turtle("square")
head.color("lime")
head.penup()
head.direction = "stop"

food = turtle.Turtle("circle")
food.color("red")
food.penup()
food.goto(0, 100)

def go_up():    head.direction = "up"
def go_down():  head.direction = "down"
def go_left():  head.direction = "left"
def go_right(): head.direction = "right"

def move_head():
    if head.direction == "up":    head.sety(head.ycor() + STEP)
    if head.direction == "down":  head.sety(head.ycor() - STEP)
    if head.direction == "left":  head.setx(head.xcor() - STEP)
    if head.direction == "right": head.setx(head.xcor() + STEP)

def move_food():
    food.goto(random.randrange(-LIMIT, LIMIT, STEP),
              random.randrange(-LIMIT, LIMIT, STEP))

def game_loop():
    if abs(head.xcor()) > LIMIT or abs(head.ycor()) > LIMIT:
        head.goto(0, 0)
        head.direction = "stop"

    if head.distance(food) < STEP:
        move_food()

    move_head()
    screen.update()
    screen.ontimer(game_loop, 100)

screen.listen()
screen.onkeypress(go_up, "Up")
screen.onkeypress(go_down, "Down")
screen.onkeypress(go_left, "Left")
screen.onkeypress(go_right, "Right")

game_loop()
screen.mainloop()
```